

# Physical Action Effects in Dexterous Interaction Models: Measurement, Randomized Identification, and Task-Utility Prerequisites

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## Abstract

We investigate whether physical action-effect information can support dexterous decisions on an existing Inspire-hand simulation substrate. Initial physical, contrast-learning and actuation-factorization screens fail their fixed gates. Cold replay acquires interacting states but fails a strict pre-intervention matching requirement. Prospective randomized assignment instead identifies conditional-effect risk differences without asserting solver-state clones. Archived predictors show no robust causal advantage on 3072 new windows; a directly supervised contrast learner also fails on 3072 further windows. Exploratory evidence from its factual control motivates a fresh randomized closed-loop test with 3072 complete episodes. Greedy selection raises mean object height by 2.60 mm versus the native actor, but retained success is 0.80% versus 1.17%, failing all utility gates. An independent label audit finds that reference lift intervals last only 25-36 frames and end in return to the table, while evaluation demands 45-step holding and full-episode nonreturn. We retain this design limitation and separately construct an explicitly synthetic 90-frame holding plateau. Existing actors retain only 5 of 768 episodes for 75 steps; one fixed 460800-interaction continuation retains 0 of 384 fresh episodes. Static elevated-pose PD targets fail the necessary retention condition in 192 trajectories. A tabletop-axis measurement error is preserved and independently corrected from saved full poses. A prospective four-dose finger-closure screen produces 0/0/6/13 retained trajectories out of 48 per dose, but no dose passes its full multi-motion gate. A subsequent randomized support-removal witness separates continued holding from release on a preselected motion in both fresh seeds; 64 trajectories are held before intervention, controls remain elevated and releases return to the table. A scratch observation-policy learner nevertheless achieves 0/192 sustained holds, as does one fixed on-policy aggregation. A separately trained reference-relative absolute-target actor holds 56/192, including 56/64 on the preselected motion, without a teacher fallback. These exploratory results establish neither a new general modeling method nor manipulation-policy improvement or journal readiness.

## 1. Problem and relation to prior work

An interaction model can fit common motion while failing to compare alternative actions. Useful comparisons additionally require experiments capable of measuring physical differences at the states in which actions are selected. We examine this prerequisite alongside bounded policy-initializer checks. The repository provides self-trained actors, geometric models and a DExplore simulation substrate [xu2025]. Historical policy audits are motivation rather than fresh evidence for this manuscript.

Shared particle actions and cross-embodiment world models already support planning [he2025]. Value-aware objectives [farahmand2017], predictor-level action recovery [qiu2026] and multi-step controllability representations [rudolph2024] are established. Contact-consistent replay and actuator residuals are also developed by Contact-UAN [fey2026]. Our small physical MLP studies do not reproduce these systems or establish priority for their general principles. The present contribution is an exploratory evidence record with frozen hypotheses and retained failures. Recent tactile world models already forecast contact evolution: DexTacWAM models joint visual and tactile futures [yuan2026], while DexTouch-WM aligns human and robot touch/action spaces and studies synthetic policy trajectories [qin2026]. WHIRL predicts future human-intervention probability and shapes actor risk [yin2026]. Generic contact forecasting or predictive risk shaping therefore cannot provide a new claim for this project. These primary sources are literature context, not reproduced baselines. WorldSimProbe already formalizes action realization and grounded environment response requirements through controlled simulator evaluation [co2026]. Facet-0 combines action-wrench prediction, a contact-aware critic and reinforcement learning adaptation [deng2026]. Generic physical grounding diagnostics or valuing contact consequences therefore cannot constitute our novelty. Our witness and baseline-learning tests identify experimental prerequisites, not a competing foundation model. Full methodological comparisons remain necessary.

## 2. Initial physical and contrast-learning screens

Four panels combine actor seeds 286/287 with environment seeds 288/289, each with 32 environments for each of three airplane motions. A retrospective trigger schedules one step after the first reference event where independent hand and object force proxies are active. A one-step wrist-z action change requests a plus or minus 10-mm PD target offset. This is a desired offset, not measured wrist travel. All subsequent normalized actions and RNG schedules are replayed. Two zero arms measure repeated process variation. Each object response subtracts its own actual pre-intervention position. Pulse and repeat contrasts are summarized by vector RMS.

| Steps | Pulse RMS (mm) | Repeat RMS (mm) |
|-------|----------------|-----------------|
| 1     | 1.914          | 0.000           |
| 2     | 2.572          | 0.084           |
| 5     | 7.150          | 0.101           |
| 10    | 18.827         | 0.444           |
| 20    | 75.463         | 6.081           |
| 30    | 105.792        | 9.638           |

Table 1. Initial fixed panels: pulse versus observed zero-repeat position contrasts.

At five and ten steps, contrast ratios are 70.44 and 42.36. At one step, the observed zero repeat is exactly zero; its ratio is undefined rather than made finite by an arbitrary denominator. Mixed vertical signs fail the joint physical gate. No beneficial lifting or population confidence bound follows from a large observed contrast.

The subsequent vector-learning decision retains this failure. It fits 192 actor-286 windows and tests 192 actor-287 windows using four past/current physical states and ten preset replay actions. Future observed physical states are absent from inputs. Matched two-layer 128-unit SiLU networks predict five/ten-step position responses; three seeds each receive 1000 Adam updates. Differential training adds measured plus-minus contrast loss. All normalization uses fit rows. Fixed test panels were already inspected by the physical experiment, making this reused-data exploration.

| Input/objective              | Five steps (mm) | Ten steps (mm) |
|------------------------------|-----------------|----------------|
| Factual                      | 10.121          | 26.546         |
| Differential                 | 10.114          | 26.990         |
| No pulse                     | 9.922           | 26.456         |
| Global fit mean              | 10.061          | 26.259         |
| Motion fit mean (privileged) | 10.137          | 26.340         |

Table 2. Actor-transfer contrast RMSE, millimeters; learned rows average three optimization seeds.

Differential training does not improve over factual or global fit-mean controls. Vertical sign agreement is 32.6%/31.1%, below its fixed 75% gate. Both initial screens are UNPROMISING. Increasing steps or choosing another seed is not used to salvage these exact designs. Physical acquisition took 481.9 seconds; fitting took 17.2 seconds on one RTX 3090.

## 3. Geometry and larger archived predictive screen

Full sampled current geometry places 318 of 384 old triggers within 20 mm. However, actor 287's motion 1 is entirely outside that range in both old test panels, with gaps 55-105 mm. Independent force proxies can both activate due to table contacts. The geometric shift is documented, without identifying it as the sole cause of learning failure or redefining the

old gate.

We next use two immutable, episode-disjoint historical collections, 960 complete episodes each. Nonterminal frames at step modulo 16 equal to zero yield 32902 fit and 32896 test rows. These collections have been used by previous research and are not pristine Validation data. The current-near test stratum contains 1756 frames from 639 episodes; proximity uses a frozen stride-8 surface convention.

The target is one-step object-position residual over constant velocity, expressed in the current object frame. State input includes joint angles and velocities, object velocities, gravity direction, object height, palm/tip relative positions and force proxies. Matched effect networks receive either zero action slots, native commands, exact PD-target/FK palm-tip motion, predicted realized hand motion, or measured future hand motion. The last is a privileged diagnostic. An independently fit actuator model predicts joint residuals before the learned-motion variant's FK. Effect networks have two 128-unit SiLU layers, fixed 1000 updates and three seeds. Sampling is half-near and half-all fit rows. No test-selected checkpoint is used.

| Effect input                      | Near RMSE (mm) | All RMSE (mm) |
|-----------------------------------|----------------|---------------|
| State only                        | 17.180         | 7.328         |
| Native command                    | 16.436         | 7.330         |
| Nominal PD motion                 | 14.599         | 7.093         |
| Predicted hand motion             | 15.066         | 7.206         |
| Measured future hand (privileged) | 13.994         | 7.001         |
| Passive                           | 23.400         | 7.587         |

Table 3. Archived one-step vector prediction error, seed-mean millimeters. Near and all-state strata differ.

The predicted-motion primary method gains 8.33% over command input, failing the 10% requirement despite beating state-only, passive and its geometric transport control. Its gate remains UNPROMISING. Actuator wrist-position errors are 11.75-12.07 mm near the object. Measured future hand motion includes possible contact feedback and cannot establish a deployable causal mediator.

The separately reported nominal-motion variant gains 11.18% and is better in each seed. This motivates a separately frozen fresh causal test; it does not retroactively replace the primary method. The label-free nominal no-slip transport control is inaccurate, with near RMSE 133.64 mm. The full screen took 53.64 seconds. All 30 saved held RMSE metrics were independently recomputed.

## 4. Fresh proximity acquisition and failed matching requirement

New environment seeds 490/491 are combined with the same two actors. A compact reference records up to 360 steps, stopping at the first native termination. For each environment, acquisition selects the first pre-action step at or after step 10 with five remaining nonterminal transitions, both current force proxies, and full sampled surface distance at most 20 mm. Future object outcomes never choose a trigger. Frozen assets provide 1538 hand and 1024 object points.

All 384 environments acquire a qualifying state, including all three motions in every panel. The maximum sampled gap per panel is 16.60, 15.57, 18.83, 18.12 mm. Eight cold processes per panel implement two zeros and one-step plus/minus pulses in world x, y, z. Initial tensors, physical properties, actor/RMS states, actions and RNG schedules are checked. Warm solver state is not asserted to be cloneable. The existing effect models are frozen before acquisition; no refit is allowed.

The intended comparison requires pre-intervention position RMS at most 0.05 mm, maximum joint difference at most 0.0001, and maximum quaternion-component difference at most 0.0001. Joint coordinates combine translations in meters and angles in radians; quaternion differences are dimensionless component differences.

| Panel     | Position RMS (mm) | Joint max | Quaternion max |
|-----------|-------------------|-----------|----------------|
| t286_s490 | 0.003554          | 0.0000732 | 0.0000670      |
| t286_s491 | 0.010859          | 0.0002204 | 0.0006188      |
| t287_s490 | 0.007110          | 0.0000000 | 0.0000416      |
| t287_s491 | 0.013398          | 0.0000982 | 0.0000881      |

Table 4. Maximum pre-intervention mismatch over arms relative to zero-a. Limits: position 0.05 mm; joint and quaternion components 0.0001.

All 32 physical arms complete, but seven arms of panel actor 286/seed 491 fail the joint/orientation precondition. The first detected failure is zero-b, before any action intervention. Execution stops with FAILED; the scientific transfer screen is UNCLEAR due to invalid matching. No favorable arm or window is retained, no threshold is widened, and no pooled causal score or model winner is reported. Acquisition and the failed analysis took 1152.21 seconds. This is a measurement blocker, not a refutation of the nominal predictor.

## 5. Exact-null recovery mechanism screen

Six native commands are overwritten by dependent finger targets. Randomizing them leaves every PD target exactly unchanged; all 32896 held rows verify this bitwise. We test whether recovering such independent null commands from model-generated object predictions induces false action responses. A truthful physical transition cannot recover an independent normalized null coordinate with population MSE below one. This is an elementary conditional-variance bound, not a novel theorem. Mutual-information sufficiency limits are already studied [rakelly2021].

Every minibatch independently randomizes null commands while keeping physical labels fixed. Four matched physical predictors receive factual loss only, full inverse loss, effective-command inverse loss, or hard-canonicalized forward input with effective inverse loss. Inverse heads take current state and generated object residuals, never actual future state. Seeds 511-513 each train for 1000 updates; the auxiliary coefficient is fixed at one. Hard canonicalization guarantees null invariance by construction, without establishing learned causal correctness.

| Variant                 | Near RMSE (mm) | Null response (mm) | Null inverse MSE |
|-------------------------|----------------|--------------------|------------------|
| Factual only            | 16.557         | 1.118              | not trained      |
| Full inverse            | 16.534         | 1.079              | 1.002            |
| Effective inverse       | 16.531         | 1.069              | 1.023            |
| Hard quotient + inverse | 16.564         | 0.000              | 1.023            |

Table 5. Exact-null intervention diagnostics, near stratum and seed means. Untrained null-head coordinates in effective/quotient variants are diagnostic only.

Full inverse null error remains near the population floor and its null response is below factual-only in every seed. Three of five interference gates fail: UNPROMISING. The proposed inverse-induced null-encoding mechanism is not supported here. Hard canonicalization removes null response, with little factual-error change. This screen took 33.06 seconds; 66 saved prediction/recovery metrics were independently audited. It is neither an AD-WM reproduction nor a general rejection of inverse learning.

## 6. Prospective randomized identification

To avoid invalid cold-replay matching, we acquire a single factual transition per independently assigned intervention. Four 768-environment panels combine actors 286/287 with initial seeds 492/493, 256 environments per motion. Assignment is drawn with a private CPU generator before physics. The trigger is the first even current step at or after 10

with both force proxies and full sampled hand-object gap at most 20 mm. A zero arm or one-step plus/minus wrist pulse in world x, y, z is executed once; the actor remains closed loop. All 3072 windows complete. Realized counts per motion are 37/37/37/37/36/36/36; recorded known propensities account for the unequal count convention.

Let  $Y$  be observed one-step world position residual over current object velocity. For each wrist axis  $j$ , define  $Z_j$  as the observed residual times the difference between the inverse-propensity indicators for its plus and minus arms. Under randomization and the independent-environment potential-outcome assumption, its conditional mean is the plus-minus physical effect  $\tau_j(X)$ . For predictions  $f, g$ , the row score is  $\text{sum}(f^2 - g^2 - 2(f-g)Z)$ , summing all three output coordinates and intervention axes. Its expectation is the squared conditional-effect risk of  $f$  minus that of  $g$ . This elementary estimator identifies a difference, not absolute effect RMSE or individually observed counterfactuals. Weak numeric interactions between batched environments remain an assumption limitation; exact solver-state cloning is not asserted.

Existing nominal-motion and command checkpoints 411-413 are frozen before the trial. Each scores all six candidate commands at the same current context. Per-model losses are averaged across seeds. Descriptive intervals use 2000 bootstrap replicates, preserving shared initial-seed/environment blocks across actors within six initial-seed/motion strata.

| Comparison            | Point (mm <sup>2</sup> ) | Central95%     | Upper95% |
|-----------------------|--------------------------|----------------|----------|
| nominal minus command | 0.122                    | [-0.919,1.187] | 1.013    |
| nominal minus zero    | -0.326                   | [-1.985,1.350] | 1.124    |
| command minus zero    | -0.448                   | [-1.393,0.455] | 0.296    |

Table 6. Randomized held-window conditional-effect risk differences, mm squared; negative favors the first predictor.

All three nominal-motion gates fail: UNPROMISING. Its historical 11.18 percent factual error advantage does not establish causal action-effect superiority. The trial takes 137.84 seconds after a retained 35.85-second device-alias engineering failure. All 3072 physical windows, risk statistics and 107 input hashes pass an independent numeric audit. We stop direct control use of these archived models.

## 7. Direct randomized learning and strong controls

The 3072 observed randomized windows now become fit data. A new test uses initial seeds 494/495 and all 3072 windows; no future test outcome chooses a model. This cohort changes the assignment protocol explicitly to independent seven-arm draws with propensity 1/7, avoiding forced assignment balance. Counts are recorded as realized; no rebalancing or partial-window selection is performed.

Context comprises 63 current physical/geometry features and 18 reference commands. Six overwritten finger-command inputs are canonicalized to zero. Motion identity and observed future states are excluded. Two nuisance folds separate acquisition seeds 492/493, keeping shared actor/environment blocks together. Each nuisance model predicts  $Y$  from context, scales only its own fit fold, and predicts the other fold. Subtracting this out-of-fold response mean before constructing the inverse-propensity pseudo-contrast preserves its conditional mean. Crossfitting and randomization are established methods, not novelty claims.

For seeds 611-613, direct models have 81 inputs, 128/128 SiLU layers and nine outputs reshaped as a world contrast matrix. Factual controls receive the same context plus six pulse indicators and predict three world residual coordinates. All use Adam 0.001, weight decay 0.0001, uniform 256-row minibatches and fixed final checkpoints. Direct/factual/nuisance models get 1000 updates; the compute control gets 3000 factual updates, matching direct plus two nuisance update counts but not asserting identical FLOPs or output capacity. All scales use fit rows only. A per-seed global mean of residualized FIT contrasts and zero effect are additional strong controls.

Fresh test risk uses original observed residuals and known propensities, without nuisance fitting on test. The gate requires a one-sided 95 percent upper risk bound below zero against every control and favorable direct-minus-factual point in every seed. Bootstrap preserves the same six paired strata as above.

| Direct minus control    | Point (mm <sup>2</sup> ) | Central95%      | Upper95% |
|-------------------------|--------------------------|-----------------|----------|
| factual                 | 23.977                   | [2.024,48.106]  | 44.201   |
| factual compute matched | 24.278                   | [2.370,48.434]  | 44.586   |
| global residualized     | 20.820                   | [-1.311,45.184] | 40.882   |
| zero                    | 20.541                   | [-1.223,45.436] | 40.638   |

Table 7. Direct randomized learner minus each frozen control, mm squared; every fixed gate fails.

Direct-minus-factual seed points 32.35/19.35/20.23 all have the wrong direction. The direct method is UNPROMISING and ends without extra updates or label clipping. Fit/test take 30.31/134.49 seconds on separate admitted GPUs, producing 3.41/11.89 MiB. Six nuisance-fold scales, 16 fit model/data files, held predictions, all risk arrays and bootstrap intervals pass independent CPU/NumPy audits.

After retaining this primary failure, exploratory comparisons between the already frozen controls suggest factual-minus-zero/global differences of -3.436/-3.158 mm squared, with one-sided 95 percent upper bounds -1.106/-1.222. More factual updates have no robust incremental advantage over 1000 updates. These comparisons were selected after primary scoring; they are candidate-selection evidence, not a successful primary trial or independent Validation. A standard factual predictor is not a new method. Its candidate value is tested below using a fresh randomized closed-loop retained-success/drop experiment.

## 8. Randomized closed-loop task test and reference compatibility

The simple factual model's exploratory risk signal motivates a new frozen task Probe, not a retuning of direct supervision. Equal-weight predictions from seeds 611-613 score seven candidate actions by one-step world vertical residual. Only a 10-mm wrist target pulse or zero is selected; invalid clipping candidates are masked. Controls are native actor, random valid candidate and global selection from seven FIT-only inverse-propensity mean vertical responses. No test outcome chooses their scores or correction dose.

Actors 286/287 with new initial seeds 496/497 provide four 768-environment panels. At reset, independently assign one of four policies with probability 1/4. Every policy gets the same current force/proximity gate, candidate inference and bounded opportunity budget: at most 10 events, at least six physical steps apart. Selected pulses last one step; the actor remains closed loop. Each environment is followed to its first native episode end. All 3072 episodes complete without selection.

Primary retained success requires 45 consecutive contact-supported steps at least 3 cm above initial object height, followed by no counted drop before native termination. A drop after primary success is height below 2 cm or lost conjunction of the two force proxies for six steps. The gate requires at least 3 percentage points retained gain and positive one-sided 95 percent lower bound against all three controls, plus upper 95 percent drop increase at most 2 points against actor. Intervals resample shared environment IDs across actors within six fixed strata.

| Policy              | Retained (%) | Stable (%) | Drop (%) | Max hold (s) |
|---------------------|--------------|------------|----------|--------------|
| Actor               | 1.166        | 11.509     | 10.342   | 0.652        |
| Random              | 1.332        | 10.588     | 9.256    | 0.675        |
| Global              | 0.542        | 9.624      | 9.082    | 0.543        |
| Conditional factual | 0.803        | 14.004     | 13.202   | 0.815        |

Table 8. Randomized complete first episodes, six-stratum policy means. Stable success is 45 steps; retained additionally excludes later counted drops.

All four gates fail: UNPROMISING. Conditional-minus-actor retained difference is -0.364 percentage points, central 95 percent [-1.287, +0.575], and drop increase +2.859 points, upper 95 percent+5.478. Secondary stable-success difference+2.496 points and mean-lift difference+2.604 mm do not replace retained success. The trial takes 465.87 seconds and 44.13 MiB on one admitted GPU. Compact trajectories independently reconstruct all 3072 labels; 30440 policy choices and 142 protected input hashes pass closeout audit. No manipulation utility claim is upgraded.

Followup exposes a task-specification limitation that should have been audited before freezing the controller gate. Raw reference labels encode short periodic lifts and return to the table at the end of every motion. Full-episode nonreturn therefore asks the reference-tracking actor to depart from its intended behavior.

| Reference               | Longest lift (steps) | Duration (s) | Final rise (mm) |
|-------------------------|----------------------|--------------|-----------------|
| s3_airplane_lift        | 36                   | 1.200        | 0.495           |
| s7_airplane_lift_Retake | 28                   | 0.933        | 0.139           |
| s9_airplane_lift        | 25                   | 0.833        | 0.160           |

Table 9. Immutable source reference labels. Every maximum continuous lift of at least 3 cm is shorter than the 45-step physical success standard.

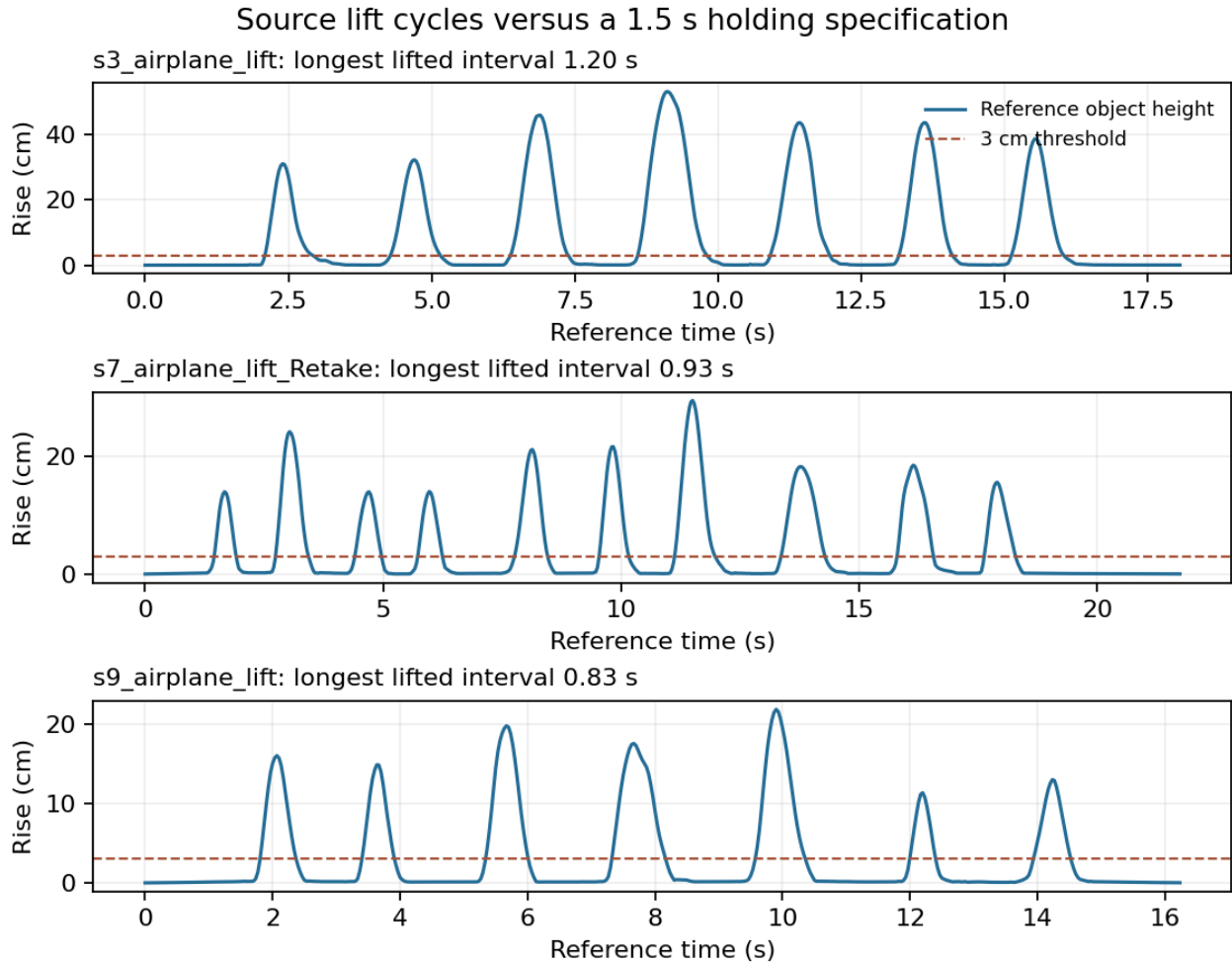


Figure 1. Source object-height targets over reference time. These are labels, not successful physical rollouts. Repeated lift/return cycles are incompatible with requiring perfect tracking and uninterrupted lift lasting 1.5 seconds simultaneously.

Eight historical native runs also have no retained success despite 70 stable instances, including correlated repeats; the new trial has small positive retained rates. Reference return does not prove the cause of every early drop. The old gate remains failed. The experiment cannot establish that all predictive control is useless or that every counted return is

accidental object loss. Repeated corrected states also differ from the first-contact contexts used to fit this model. Both task specification and closed-loop distribution need attention.

Next we freeze an explicitly synthetic holding task: duplicate 90 raw reference rows at the maximum height of each FIRST lifted interval, without selecting a pose using simulator outcomes. The native loader recomputes velocities. A new actor-only feasibility test asks for 75 consecutive contact-supported physical steps in that inserted phase, before deliberate release. Generation preserves initial rows and original remainder; it is not human-data or task-transfer evidence. Completed results of the new task are reported separately below.

## 9. Synthetic holding transfer and one frozen curriculum

The three first-lift label peaks are frames 72/50/62. Inserting 90 identical raw rows after each peak creates stationary phases 73-162, 51-140, 63-152, and new reference lengths 633/743/578. All 598 columns are copied together; original initial rows and remainder are preserved. The native loader recomputes all velocity channels, independently verified zero throughout the plateau.

A separate feasibility gate requires 75 consecutive physical steps within the 90-row phase with root height at least 3 cm above its actual initial height and both force proxies. It requires at least 10 percent pooled success and 5 percent in every motion. Actual native progress, rather than collector-loop indices, defines phase membership. Reference length and the complete first-episode endpoint are independently checked; no early native episode is selected.

Actors 286/287 with fresh seeds 498/499 provide four 192-environment panels, 64 per motion per panel. All 768 native episodes complete with actor/RMS frozen. The original parent reporting stage fails on a duplicate keyword; physical files are complete and preserved. A separate version uses the same frozen scoring functions and independently reconstructs every label, without rerunning physics. All 156 protected input hashes remain unchanged. Only 5/768 retained75 and 19/768 stable45 events occur; all four gates fail, UNPROMISING.

| Screen           | Motion | Episodes | Retained75 | Stable45 | Rise (mm) |
|------------------|--------|----------|------------|----------|-----------|
| Actor transfer   | 0      | 256      | 4          | 6        | -10.025   |
| Actor transfer   | 1      | 256      | 0          | 0        | 5.149     |
| Actor transfer   | 2      | 256      | 1          | 13       | 14.190    |
| Actor transfer   | Pooled | 768      | 5          | 19       | 3.105     |
| PPO continuation | 0      | 128      | 0          | 0        | -1.703    |
| PPO continuation | 1      | 128      | 0          | 0        | 3.906     |
| PPO continuation | 2      | 128      | 0          | 0        | -1.396    |
| PPO continuation | Pooled | 384      | 0          | 0        | 0.269     |

Table 10. Synthetic plateau episodes. Transfer and continuation use different cohorts and actor configurations; these rows are separate feasibility screens, not a matched treatment comparison.

One separately frozen baseline continuation starts from self-trained actor 286's fixed epoch 420 network and both observation statistics, with fresh optimizer and new epoch count. Seed 721 receives exactly 300 new PPO epochs/460800 interactions: 96 environments, horizon 16, minibatch 512, four PPO passes, constant learning rate 0.00001. The final checkpoint alone is evaluated; no best/intermediate selection or additional optimization seed is used. Wrist-translation exploration standard deviation is predetermined at 5 mm; other coordinates retain their source values.

Half of training resets start at the first stationary row through epoch 80; this probability decreases linearly to zero at epoch 200. The last 100 epochs all start at frame 0. Native reference reward receives an additional bounded contact-supported lift bonus of coefficient 10 only during the holding phase. There is no Cm, action-selection teacher or other predictor. Training exposure is not success: 68947 plateau samples and 7933 positive bonus samples are privileged

diagnostics.

The final checkpoint and evaluated model/statistics fingerprints match exactly; all weights remain finite. Fresh seeds 500/501 each complete 192 first episodes. No retained75 or stable45 instance occurs in 384 episodes; the force-proxy conjunction is absent throughout the phase. All four gates fail, UNPROMISING. All 178 protected inputs and independent trajectory labels pass closeout checks. This exact recipe ends without extra epochs, seed or learning-rate tuning. The transfer and training screens take 400.79/1120.95 seconds, respectively; one failed continuation does not establish that self-training or all curricula are impossible.

## 10. Native pose semantics and mechanical grip prerequisites

The actual native reset method clamps and couples finger DOFs. A three-pose label audit compares raw reference coordinates with that method applied to dummy tensors. Maximum changes are 0.29677/0.26261/0.09957 rad. Sampled surface gaps change from 0.9135/0.9456/0.4883 mm to 0.3453/1.4025/0.8457 mm; geometric finger flags remain 3/1/1. Thus reset changes do not explain failure through a demonstrated loss of these minimum-proximity or binary-flag quantities. Unsigned proximity does not establish nonpenetration or a compressive grasp.

A mechanical diagnostic bypasses the actor: initialize at the elevated zero-velocity stationary row and hold the actual reset q as a constant absolute PD goal for 90 physical steps. Native action inversion is verified within  $2.981e-8$ ; actor invocation raises an error and model/statistics remain unchanged. Two fresh seeds 502/503 produce 192 complete trajectories. None satisfies 75 consecutive ticks of the necessary height/proxy criterion: root at least 3 cm above original frame 0 and at most 1 cm below its elevated initial height, with both force proxies. This is neither frame 0 pickup nor proof of intrinsic grasp infeasibility.

The prospectively specified full-mesh table clearance contains a measurement error: local Z was assumed the tabletop normal. The actual 21304-vertex table is thin along local Y, and its native xyzw pose maps local -Y to world +Z. Original full-clearance metrics are therefore INVALID\_TABLETOP\_AXIS and cannot support a physical-distance claim. Original code, counts and failed gates remain intact. A separately recorded offline correction uses all 25002 object vertices and the actual tabletop support plane. Initial clearances are 307.03/130.93/153.56 mm. Independent NumPy world-up support agrees within 0.000000118 m; corrected retained75 remains 0/192. This is explicitly post-hoc reused-data geometry, not prospective Validation. The independently valid necessary-condition failure does not depend on the mistaken clearance axis.

A followup saved-joint audit excludes another apparent issue: wrist rotations are URDF continuous joints with no position bounds. A helper's default plus/minus pi range is not a simulator hard limit. Motion 2 joint 5 at -4.59205 rad is tracked near that value. All genuinely bounded goals satisfy their specified bounds. Nominal PD-goal agreement must also be distinguished from realized tracking: maximal wrist position deviations are 26.903/0.351/0.132 mm and finger deviations 0.0661/0.1940/0.0693 rad. These observations do not identify the cause of grip failure.

A new prospective mechanical screen asks whether additional finger closure can supply holding forces while keeping the wrist fixed. Doses 0/0.05/0.15/0.30 rad are added to independent curl joints 6/8/10/12/15; thumb yaw 14 and wrist 0-5 are unchanged. Each parent is capped by both its own and dependent native limits, then dependent coupling is reconstructed. Initial physical q is unchanged; closure is an actuator target at the first step, not a state teleport. Six overwritten command coordinates remain canonical zero. No policy is invoked and no action is clipped.

Fresh seeds 504/505 each use 96 environments. Within each motion a private generator randomly permutes eight copies of each dose before physics, giving 16 trajectories per dose/motion overall. All 192 run 90 physical steps. The full retention criterion is prospectively implemented with the correct thin tabletop geometry and independently reconstructed world-up clearance, force proxies and longest runs. Each positive dose must reach 50 percent pooled, 25 percent in every motion, and at least 25 percentage points pooled gain over zero. Candidate dose screening is not formal Validation.

| <b>Dose (rad)</b> | <b>Pooled /48</b> | <b>Motion0 /16</b> | <b>Motion1 /16</b> | <b>Motion2 /16</b> |
|-------------------|-------------------|--------------------|--------------------|--------------------|
| 0.00              | 0                 | 0                  | 0                  | 0                  |
| 0.05              | 0                 | 0                  | 0                  | 0                  |
| 0.15              | 6                 | 6                  | 0                  | 0                  |
| 0.30              | 13                | 8                  | 0                  | 5                  |

*Table 11. Prospective bounded closure screen: retained 75 counts at each dose. All dose gates fail; motion-specific counts are retained rather than replacing the pooled and every-motion criteria.*

No dose meets all gates, UNPROMISING. The 0.30 rad dose meets the gain requirement, but its 27.08 percent pooled rate and zero success on motion 1 fail the other conditions. Successful instances on motions 0/2 expose a concrete mechanical candidate signal, without proving pickup, policy learning, force closure or generalization. This exact four-dose family ends without larger-dose or selected-seed expansion. The screen takes 76.99 seconds and 9.43 MiB; all 189 protected hashes remain unchanged. Independent clearance disagreement is at most 0.000000482 m. The distinct next question is grasp-pose construction and contact geometry, before another baseline learner. These statements describe the decision at that screen; the subsequent frame 0 witness and independent learners appear below.

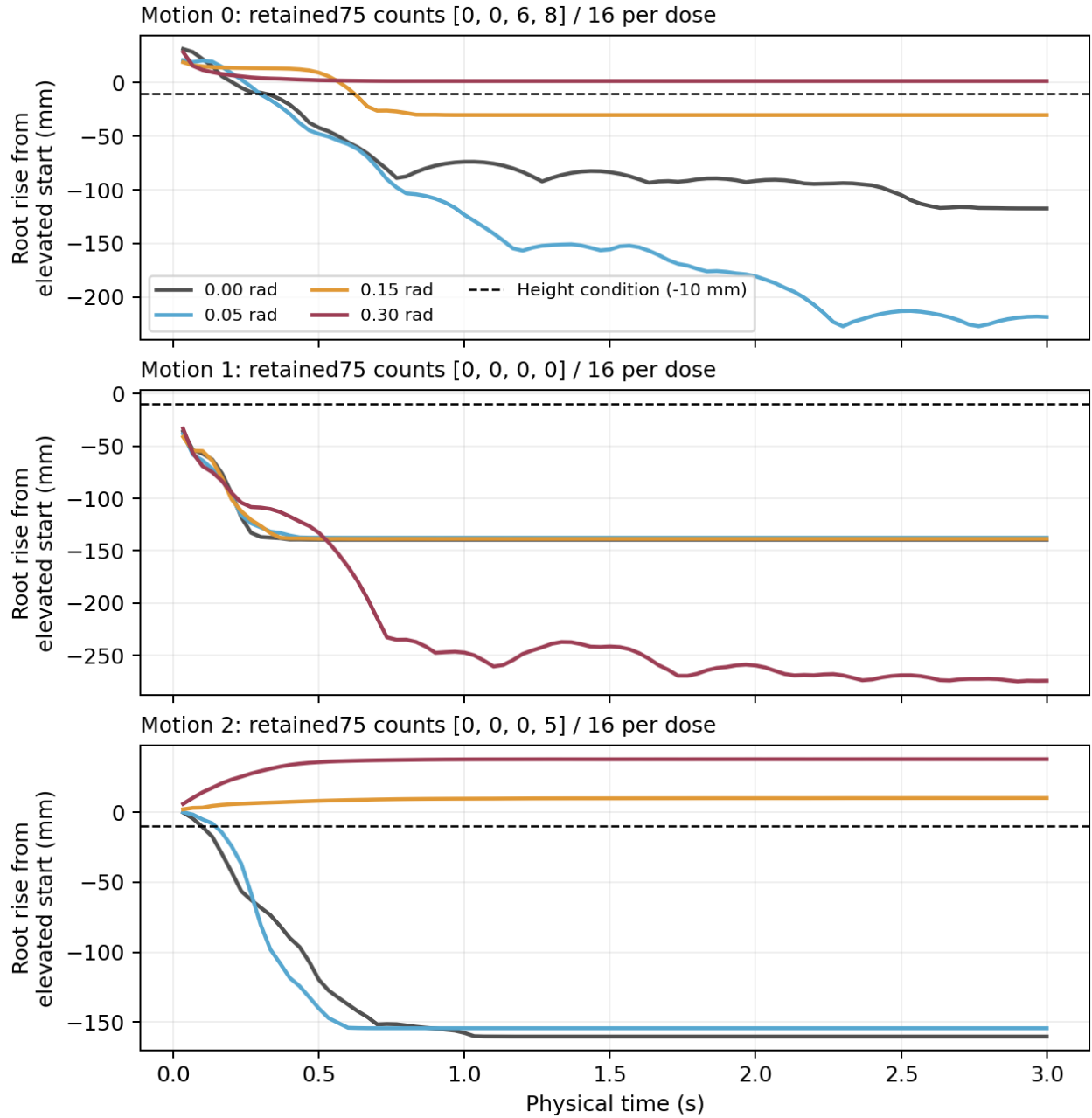


Figure 2. Physical object-root trajectories for the prospective closure screen. Each curve averages 16 trajectories per motion/dose. The dashed line is only the initial-height condition; forces, table clearance and 75 consecutive steps are additionally required. Mean height is not retained success or a policy benefit.

## 11. Prospective support-removal witness

An original-frame 0 tracking controller follows the available reference joint plan, with a timed 0.30 rad curl increment, then holds the last plateau target. Its original strict-force gate remains UNPROMISING: unchanged/closure counts 0/96 versus 5/96 on fresh 506/507. Geometry-only motion 1 retention 32/32 is explicitly secondary. The actual object weighs 2.5936 g under native gravity; the 0.1 N net-body force threshold is 3.93 times its weight. Neither this ratio nor geometry alone establishes the hand's contribution, motivating a prospective removal experiment.

Before new physics, motion 1 is selected as the primary stratum. In each fresh seed 508/509, 96 environments cover all three motions with 32 each. Private random assignment balances 16 continued-hold and 16 release environments per motion. All follow the same controller through the plateau stop. Release then commands open fingers and 20 cm wrist retreat; controls retain their last target. All 202 physical ticks run with gravity, unmodified object state and three native

actors. The horizon is corrected from 200 to 202 before physics to cover every motion's late window; primary motion 1 gates are unchanged. Warm solver-state clones are not asserted. Five requirements per seed are fixed: at least 80 percent prior geometry75 in each arm, 90 percent realized retreat of 15 cm, 80 percent control late retention, 80 percent release late loss, and 30 mm mean height separation. The late window is stop+30 through stop+40 inclusive, all 11 ticks required.

| Seed | Prior hold | Prior release | Late hold | Late loss | Height (mm) |
|------|------------|---------------|-----------|-----------|-------------|
| 508  | 16         | 16            | 16        | 16        | 147.113     |
| 509  | 16         | 16            | 16        | 16        | 144.379     |

Table 12. Prospective primary-motion support witness. Counts retain all 16 environments per arm and seed. Prior geometry75 precedes intervention; late loss/retention require all 11 late ticks. Height is control-minus-release mean in the late window.

All ten predeclared gates pass, PROMISING as a mechanical Probe. Every release achieves measured retreat; control late retention and release late loss are 32/32. Motion 0/2 remain unsuccessful in both arms and are not excluded from the archive. Historical strictforce75 counts are 47/64 on the 64 prior geometrically held primary trajectories. This identifies disagreement for this verified setting, not a universal force-proxy error rate. The intervention provides evidence of mechanical support, not force closure or learned-policy competence. Runtime 102.35 s, 17.03 MiB, all 193 protected inputs unchanged; independent full mesh clearance error  $3.80\text{e-}8$  m.

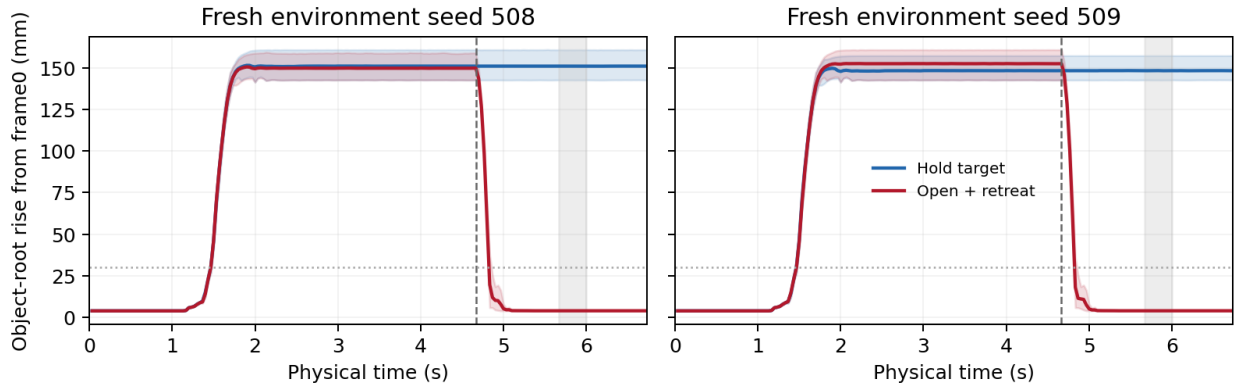


Figure 3. Actual primary-motion object-root trajectories, all 16 assigned rows per curve. Bands show minimum/maximum across trajectories, not confidence intervals. Dashed vertical line marks intervention; shading marks the 11-tick late window. No learned actor controls these trajectories.

## 12. Independent observation-policy learning

The witness motivates a new prospective holding criterion: object-root rise from actual frame 0 at least 30 mm and full mesh tabletop clearance at least 20 mm at every tick from plateau-stop-74 through stop+30 inclusive. This 105-tick window includes 75 plateau ticks followed by 30 drop-checking ticks under the holding reference. It does not relabel any historical failed gate. Strictforce 75 remains secondary. Known motion 1 must reach 50 percent pooled and 25 percent in each fresh seed; all three motions remain in training and reporting.

A random 512/256/128 ReLU/Tanh policy uses 70 features: current native joint position and velocity, current object root state, two current force proxies, 18 planned joint coordinates and planned phase fraction. Both teacher and actor have the planned reference; no observed future state, teacher action or outcome enters the actor. Seed 510 supplies all 19392 teacher rows. Fixed 2000 Adam updates, lr 0.001, batch 512, weighted action errors and final-only checkpoint use FIT normalization. The six hardware-null command coordinates are canonicalized. No source actor weights are used; it is loaded solely for native environment construction and never called. New 511/512 evaluations contain 192 complete 202-tick trajectories without fallback. Physical 105 and strictforce75 are both 0/192, UNPROMISING. Training loss is not manipulation success. Causal input and native PD reconstruction independently pass.

A separate descriptive saved-data diagnostic finds motion 1 late wrist-target RMSE 170.58/171.36 mm; input normalization does not clip any motion 1 policy context. Independent teacher-query reconstruction matches saved teacher targets. These observations motivate, without establishing a causal explanation, one fixed on-policy aggregation experiment. Fresh 513/514 collect only frozen-policy actions; teacher labels are queried offline at visited states and never executed in these cohorts. All original 19392 teacher rows and 38784 visited-state rows are retained. No 511/512 test row is fitted. Baseline weights and normalization are preserved, optimizer reset; sampling seed 735, 2000 updates and all other hyperparameters fixed. Final-only evaluation uses new 515/516 seeds and the same physical105 gate. The aggregation also fails both gates: physical105 and strictforce75 are 0/192, including 0/64 on motion 1. The 184.70 s run preserves 217 protected inputs; this exact aggregation ends without further rounds. Smaller posthoc wrist-target errors do not replace the failed manipulation gate.

A distinct scratch actor changes the learned output to reference-relative absolute PD targets. Residual scales are 20 mm for wrist translation, 0.10 rad for wrist rotation and 0.40 rad for fingers. Targets enforce native bounds and coupling; the exact native PD inverse supplies feedback from actual current joints. Original 510 teacher rows alone train the 70-input network, seed 736, 2000 updates, unit residual MSE and final-only evaluation. No timed curl schedule executes in testing. New 517/518 produce 56/64physical105 on primary motion 1, 28/32 each seed, passing both initializer gates: PROMISING. Motions 0/2 each remain 0/64; pooled 56/192 and historical strictforce75 is 50/192. The 117.16 s run verifies all 226 protected inputs, causal context and independent residual-target/PD reconstruction. This provides an initializer, not a reference-free actor, full multi-motion competence or Cm benefit. Different losses, coordinates and physical cohorts prevent a matched superiority claim over prior learners. All earlier failures remain failed; baseline-only tuning stops here.

| Policy          | Motion | Trajectories | Physical105 | Force75 |
|-----------------|--------|--------------|-------------|---------|
| Scratch BC      | 0      | 64           | 0           | 0       |
| Scratch BC      | 1      | 64           | 0           | 0       |
| Scratch BC      | 2      | 64           | 0           | 0       |
| Scratch BC      | Pooled | 192          | 0           | 0       |
| One aggregation | 0      | 64           | 0           | 0       |
| One aggregation | 1      | 64           | 0           | 0       |
| One aggregation | 2      | 64           | 0           | 0       |
| One aggregation | Pooled | 192          | 0           | 0       |
| Absolute target | 0      | 64           | 0           | 0       |
| Absolute target | 1      | 64           | 56          | 50      |
| Absolute target | 2      | 64           | 0           | 0       |
| Absolute target | Pooled | 192          | 56          | 50      |

Table 13. Independent learned-policy evaluations: physical105 is the new prospective primary and strictforce75 is a historical secondary. All variants retain all three motions. These are initializer Probes, without Cm or policy-training utility claims.

## 13. Limitations and research decision

These experiments involve one object, three shared motions, two existing actors and small fixed fitting procedures. Contact proxies plus sampled proximity are not attributed pairwise forces. Short-horizon measurement is not stable lift, retention or manipulation utility. One fixed baseline PPO continuation is presented; matched model-guided policy training, unseen-object/hand evaluation and hardware deployment remain absent. Optimization seeds do not create independent physical datasets. Existing papers' claims are not replaced by these diagnostic tests.

The direct learner and tested greedy controller are stopped at their fixed gates. The new reference-target initializer supports a prospective model-assisted policy-training comparison on the defined holding task. A compatible reference is a prerequisite, not proof of controllable contact or model benefit. Synthetic plateau transfer, one fixed continuation and mechanical feasibility screens are completed with their original failed gates retained. Corrected geometry is labeled explicitly as a separate reused-data diagnostic.

The original user objective remains a journal-level research outcome. This working paper does not meet it: a distinctive method and consequential matched task evidence remain necessary. Negative outcomes and the unresolved causal screen are retained to guide further work rather than converted into positive conclusions.

## Reproducibility

All new work is isolated in branch agent/contact-response-cm. Original checkpoints, motion assets and prior evidence are read-only. Physical/differential code is fixed at 466e80d/44f22ea; factorization at bbd5ba4; fresh collection at 59de1cf; null recovery at 33d1085. Randomized collection uses 510a00e; direct fitting 2075314; independent auditing f283bb9. Closed-loop task collection uses c0ec955; task/reference audit and next synthetic task design use b2b26e0. Holding collection/corrected reporting use 40e73f2/3f4ae77; the frozen continuation uses e770d4b. Native reset/static diagnostics use da39660/1e38cf6; the independent geometry correction uses 68bf178; prospective finger closure uses 5d7c375. Frame 0 tracking uses d15ef0f, support witness 49cf8b3, scratch observation policy 9c0a218, and fresh corrective-label aggregation e2a38c2, reference-target actor 596c5a0. Manifests preserve commands, input SHA, GPU UUID, budgets and failures. Every table and figure comes from recorded labels or physical trajectories, clearly distinguished. The PDF is a ReportLab review copy; portable LaTeX source and tables are also retained. Earlier manuscript revisions are preserved separately.

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